Embodiment Shapes Rolling Behavior in a Multimodal Infant Model

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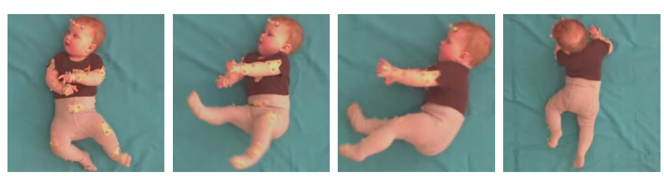
***Abstract*— Rolling over is one of the earliest milestones in infant motor development, reflecting the emergence of coordinated, whole-body sensorimotor control. Here, we conduct a computational study of infant rolling using MIMo, a virtual infant embodiment equipped with proprioception and vestibular sensation. MIMo learns both supine-to-prone and prone-to-supine rolls with reinforcement learning. Interestingly, the learned behaviors capture developmental trends and coordination patterns consistent with those reported in real infants, including improved performance and faster execution with age. Our results explain how infant capabilities and constraints can give rise to realistic behaviors in artificial agents, with a particular emphasis on how motor development is shaped by the changing body morphology. This work highlights the role of embodied computational models as a powerful tool for studying sensorimotor development.**

I. INTRODUCTION

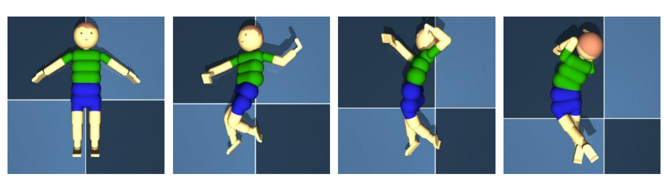
Early motor development is characterized by the emergence of sensorimotor behaviors in a rapidly growing body [1], [2]. Rolling is one of the first major motor milestones, requiring the coordinated control of the whole body. Infants learn to roll between 2 and 7 months of age [3], once they can lift their heads in the prone position and before sitting and crawling. Rolling has been described extensively in observational studies [4], [5], [6]. Common analysis techniques include the classification of movement patterns in infants [4] and in adults [7], ground pressure analysis [8], and studies on the effect of the mechanical environment, such as rolling on an incline [9], [10]. However, the mechanisms underlying the emergence of rolling remain difficult to isolate due to experimental limitations with infant studies. By way of example, Siegel et al. [11] have performed a challenging study of the muscle activations involved in roll overs using electromyography (EMG). However, they only measured the muscles from the legs and torso, but not arms and head.

Some of these experimental limitations can be overcome by making use of computational models as a complementary methodology. A promising approach is to rely on models and artificial agents that can explain the interplay between different components of development beyond the scope of experimental studies. In particular, embodied agents with realistic physics simulations offer an interesting framework for learning and analyzing behaviors. To the best of our knowledge, there are no existing computational models recreating infant rolling in artificial agents, although roll overs have been considered as a first milestone in a locomotion curriculum for simulated robots [12]. On the other hand, humanoid robots have been used to explore rolling patterns on pressure sensitive mats [13] as well as motion kinematics [14]. However, little attention has been given to the role of the embodiment in the learning of this behavior.

In this work, we present a computational model of infant rolling using MIMo [15], a multimodal embodied infant model (see Fig. 1). Our model combines a physically realistic body with proprioceptive and vestibular sensing. Rolling behavior is actively learned through reinforcement learning, allowing MIMo to autonomously discover how to roll given his bodily constraints. We investigate two key aspects of rolling. First, we explore the developmental embodiment progression by varying MIMo's body between the ages of 1 and 9 months while keeping all other training components unchanged. Second, we analyze the resulting behaviors using metrics proposed in experimental psychology, including coordination patterns and muscle activations. Our work highlights the potential of embodied computational models as complementary tools for studying sensorimotor development.



(a) 8-month-old infant.



(b) 9-month-old MIMo [15].

Fig. 1: Supine-to-prone rolling in real and simulated infants.

This work was supported by the Deutsche Forschungsgemeinschaft (German Research Foundation, DFG) under Germany's Excellence Strategy (EXC 3066/1 "The Adaptive Mind", Project No. 533717223). J.T. was supported by the Johanna Quandt foundation. We thank Miles Lenz for his work on an earlier version of the rolling simulations and Matej Hoffmann for granting permission to include the frames of the infant from Fig. 1.

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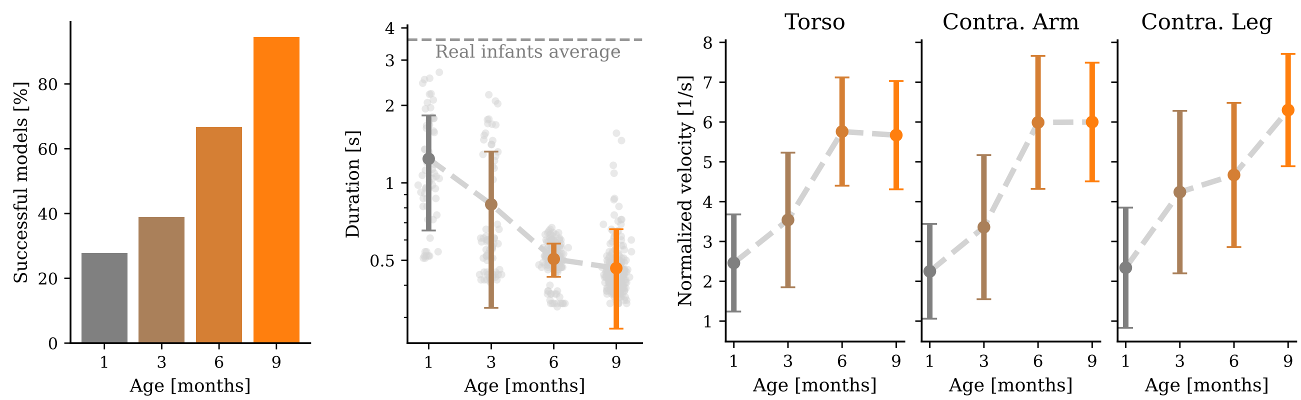


(a) Success rates.

(b) Durations.

(c) Normalized velocities.

Fig. 2: Supine-to-prone rolling statistics for embodiments of different ages. (a) Success rates are measured over 40 test episodes. A model is deemed successful if it achieves a roll over in at least 75% of those episodes, i.e., 30 out of 40. (b) Roll durations are measured from the start of an episode until the success condition of Eq. 2 is met. The average real infant duration is the average value reported by [11]. (c) Normalized velocities are computed by normalizing the total displacement of each limb from the beginning to end of the roll, facilitating comparisons across different body parts.

II. METHODS

A. Environment

We use MIMo [15], a virtual infant model designed to capture fundamental aspects of infant sensorimotor development. MIMo's age can be varied between birth and 2 years [16]. MIMo is physically grounded in the MuJoCo simulation platform, which allows us to simulate the consequences of body contacts, gravity, and movement. MIMo is equipped with multimodal sensing, in particular proprioception and a vestibular system. Proprioception provides information about the current joint angles, velocities, torques, and control commands, while vestibular sensation measures the orientation and motion of the head. MIMo can actuate each of his joints, amounting to a total of 46 degrees-of-freedom (DOF), with a spring-damper model.

The environment includes MIMo lying on a flat surface. Supine-to-prone rolls are analyzed most throughout the literature [4], [5], [11], [7] and thus are our main focus here. The emergence of rolling happens between 2 and 7 months [3], experimental studies typically study infants around the ages of 6 and 9 months [4], [8], [11]. To match this and also explore the developmental trajectories, we use 4 different embodiment ages: 1, 3, 6, and 9 months.

B. Training

MIMo learns to roll over through reinforcement learning. He must learn a policy that maps the multimodal sensory inputs to continuous commands for all joints. To do so, he is trained to maximize an extrinsic densely-shaped reward function [17] and a large sparse reward upon rolling success (see Section II-D) with Proximal Policy Optimization (PPO) [18]. We use the default implementation from the Stable-Baselines3 library [19].

Each training episode lasts a maximum of 500 steps, each step consisting of a physics simulation of 10 ms, with early stopping if the roll over is successful. The total training time consists of 10^6 steps (corresponding to under 3 hours of practice). By default, we use a learning rate of $3 \cdot 10^{-4}$, with different values compared in the Appendix. All the training procedures are repeated with multiple random seeds.

C. Roll progression

To evaluate the transition from supine to prone, we use a metric based on the orientation of the torso and pelvis relative to the direction of gravity. In the supine position, both the torso and pelvis face upwards, opposite to the direction of gravity. In the prone position, the opposite is true. We measure the achieved shift of orientation from the starting position to the goal position. In a supine-to-prone roll, the goal is always aligned with the direction of gravity. To quantify this, we compute the dot product of the local sagittal axes (the axes pointing forward) between pelvis and torso and the global gravitational axis. This dot product gives us a continuous value $\rho_i \in [-1, 1]$ ($i \in \{\text{pelvis}, \text{torso}\}$).

Finally, we average both to obtain a single orientation metric:

$$\bar{\rho} = \frac{\rho_{\text{pelvis}} + \rho_{\text{torso}}}{2}. \tag{1}$$

Additionally, we use $\bar{\rho} \geq 0.95$ as a boolean condition for a successful roll. A half-roll, in which a baby lies on its side, is identified with the condition $\bar{\rho} \geq 0.5$.

D. Reward function

A successful roll yields a large sparse reward. However, sparse rewards are notoriously challenging in large action spaces, such as the roll over environment used here. To address this, we use a densely-shaped reward function [17] generated with Potential Based Reward Shaping (PBRS) [20]. We use the potential function $\Phi(s) = \bar{\rho}$, and scale the potential difference by a factor of 100. Finally, we incorporate metabolic costs that penalize MIMo for excessive

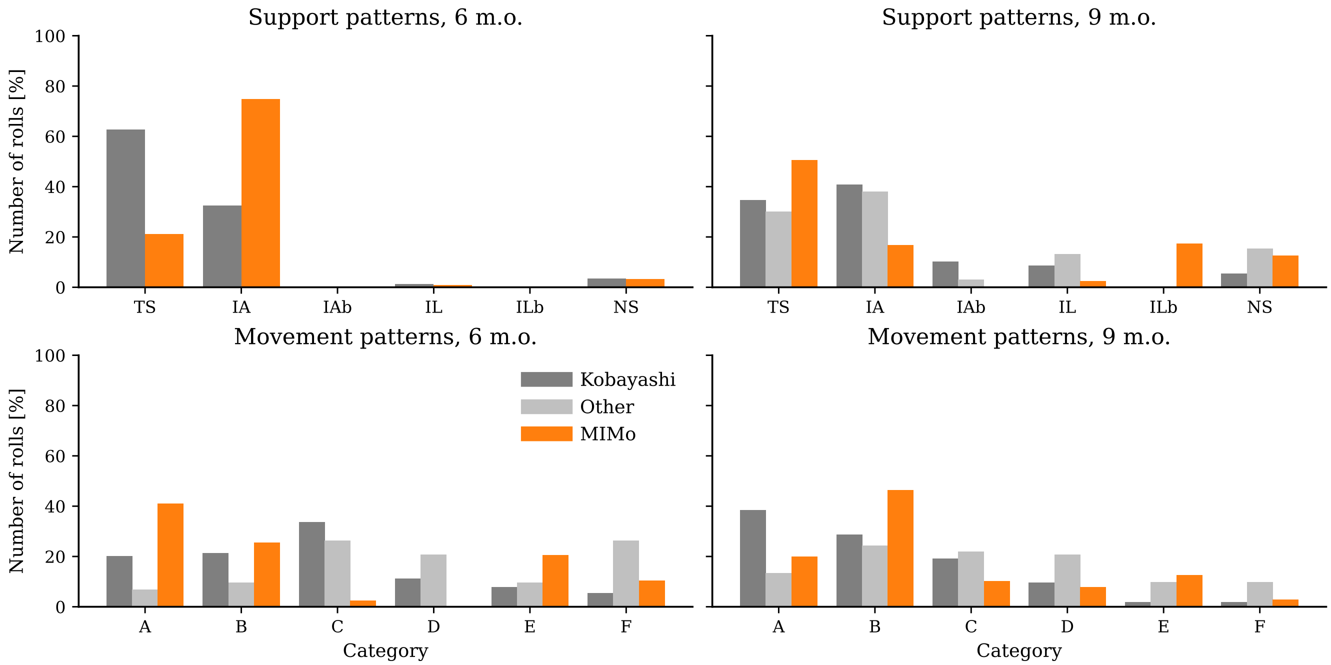


Fig. 3: Distribution of support (top) and movement (bottom) patterns displayed by "younger" (left) and "older" (right) infants. Real infant data is taken from different sources. In dark gray we show the results from Kobayashi et al. [4], who measure both coordination patterns for both age groups. In light gray are the distributions reported in [8] for "older" infants (both patterns) and in [11] for "younger" infants (movement patterns only). See Appendix for details about the patterns.

usage of his actuators. This penalization is computed as the squared sum of the control values c and scaled by a factor $\alpha \geq 0$. Put together, the reward function is expressed as:

$$r_t = \begin{cases} 500 & \text{if } \bar{\rho} \geq 0.95 \\ 100 \cdot (\overline{\rho_{t+1}} - \overline{\rho_t}) - \alpha \sum c^2 & \text{otherwise} \end{cases} \tag{2}$$

The penalization factor α is set to 0.02 by default. Different values are compared in the Appendix.

E. Evaluations

After training, the deterministic policy is frozen and used for evaluations. Unless specified otherwise, we evaluate each model over 40 test episodes and average the results. We also average across multiple runs of the same training condition with different random seeds.

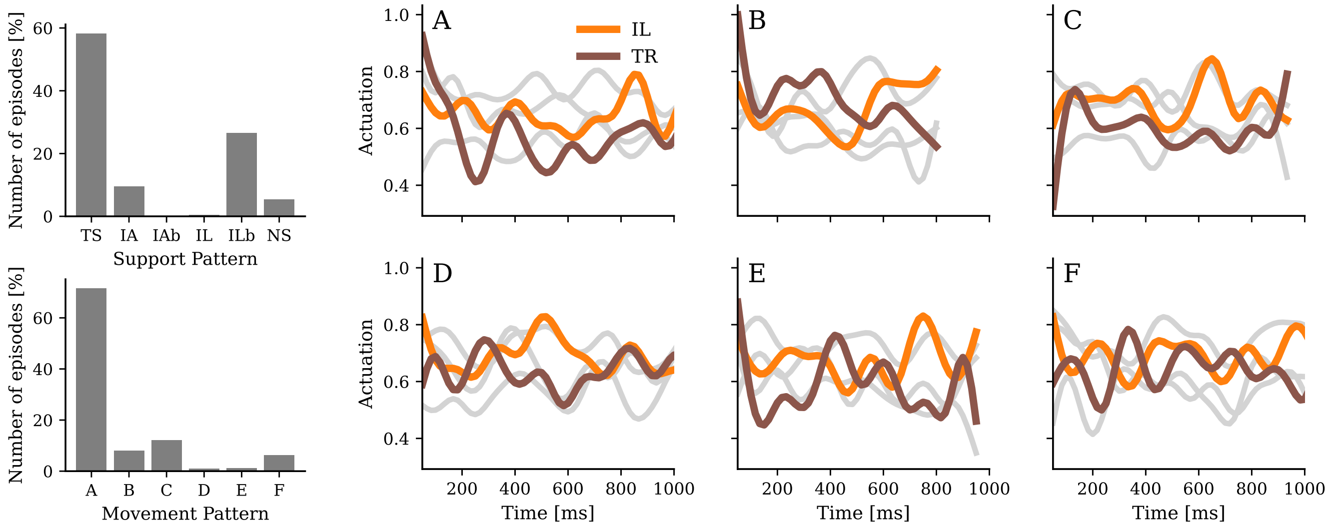
1) Success rate: A model is deemed successful if it completes a roll over in at least 75% of the 40 test episodes. The success rate is computed as the fraction of successful models per condition. Besides the success rate, we also keep track of the roll duration, defined as the time from the start of an episode until the completion of a roll over.

2) Body velocities: By using a simulator like MIMo, we can easily keep track of the movement statistics of the entire body. Here, we choose to recreate the analysis reported by Kobayashi et al. [4]. To do this, we attach MuJoCo "sites" at the wrists, ankles and torso. Contrary to the experimental counterpart, the markers are never obscured and thus there is no need for secondary markers. Velocities of each limb are calculated by differentiating the measured displacement of each marker. Since each limb must cover a different distance during a roll over, Kobayashi et al. also use a normalized velocity. They propose to fit the total displacement to a sigmoid function and normalize it to amplitude 1. The normalized velocity is then taken as the slope at the center point, in units of $1/\text{time}$. The body parts tracked are the torso (TR), ipsilateral arm (IA), contralateral arm (CA), ipsilateral leg (IL), and contralateral leg (CL).

3) Coordination patterns: There exist two different approaches to categorize the coordination patterns of a roll over (see Appendix). We adapt both to MIMo and perform the categorization of test episodes automatically following the procedures decribed next.

Stationary patterns: The first and easiest approach consists of classifying a roll over according to the resulting stationary limbs [4], [8]. To do so, Kobayashi et al. [4] calculate the time at which the torso reaches maximum displacement velocity in the direction of rolling, T_{TR}. They define a window of $250\ \mathrm{mm}$ around this time, and establish that a limb is stationary if its average velocity during that window is lower than a threshold of $100\ \mathrm{mm/s}$. We repeat this analysis with MIMo, with one critical difference. Since MIMo rolls much faster than real infants (see Fig. 2), we scale window and threshold values. The median duration of a supine-to-lateral roll for a 6-month-old MIMo is around $500\ \mathrm{ms}$, roughly 6 times faster than the values reported in [4] and [11]. Thus, we split this difference to define a shorter window size of $83\ \mathrm{ms}$ and a higher threshold of $300\ \mathrm{mm/s}$.

Movement patterns: More refined patterns can be obtained by computing the temporal dynamics of the roll over



(a) Distribution of patterns a from single trained model.

(b) Examples of actuation commands over different movement patterns (A–F). Highlighted colors correspond to the ipsilateral leg (IL) and torso (TR). Other body parts are shown in gray.

Fig. 4: Variability in coordination patterns and actuation from a single training run. (a) The model is evaluated in 1,000 test episodes with random initial conditions, and the resulting support and movement patterns are calculated for each episode. (b) Single episodes from each movement pattern are selected arbitrarily among the tests to visualize the actuation commands.

[4], [11], [8], which requires calculating the relative motion of the limbs. We adopt the methodology proposed in [4]. We fit a sigmoid function to the displacement of each moving limb to determine the time of maximum displacement velocity. This time is compared to T_{TR}. Limbs that reach their peak velocity inside a window of 83 ms around T_{TR} are classified as "synchronous", whereas limbs reaching peak velocity before or after this interval are declared as "leading" and "following", respectively. As a result, the rolling motions can be classified in 6 categories, labeled A through F, as described in the Appendix.

4) Actuation control: Finally, drawing inspiration from the work of Siegel et al. [11], we analyze MIMo's actuation control during a roll over. Instead of recording muscle activity with EMG, we directly extract the actuation control commands selected for each DOF. MIMo has a total of 46 DOF, which makes the actuation analysis complex. To simplify, the actuation time series are filtered using a 6 Hz recursive Butterworth filter and averaged across all the control commands in the kinematic chain of each body part. For this study, we focus exclusively on the torso and four limbs. Finally, we compare the temporal dynamics of the actuation commands under different conditions.

III. RESULTS

A. How embodiment constrains rolling

Infants learn to roll somewhere between the ages of 2 and 7 months [3], a point in development that is characterized by rapid changes in both body, brain, and experiences. The relative contributions of these three components are difficult to disentangle in behavioral experiments. Here, making use of a simulated infant like MIMo, we attempt to address this. To do so, we train the same deep reinforcement learning algorithm (brain) for the same amount of episodes (experience) but with four different embodiments, corresponding to 1, 3, 6, and 9-month-old.

The results are shown in Fig. 2. We find that increasingly older bodies allow MIMo to achieve higher success rates and roll faster. There are only two physical differences between these different models. First, MIMo's body grows to match an average infant of the same age, meaning a bigger torso, head, and limbs. Intuitively, it should be simpler to roll with a smaller (and thus lighter) body. However, body growth also entails an increase in strength and capabilities. López et al [16] reported that MIMo can only lift his head with a 2-month-old body and his legs with a 4-month-old body, in line with real infants. In our experiments, despite all four models being trained with the same algorithms and for the same duration, the differences in strength between the embodiments translates into large performance gaps.

These results must only be interpreted qualitatively. As seen in Fig. 2.b, MIMo is considerably faster than real infants. That is, not only because our model only has few of the many constraints and limitations of a real baby, but also because it is trained with an extrinsic reward, i.e., his only purpose is to achieve a roll over. Regardless, these experiments reveal how changes in the embodiment may facilitate or constrain this typical infant behavior.

B. Discovery of diverse successful movement patterns

Having found that MIMo can learn to roll, we inquire next whether the way in which he does so matches real infants. One important aspect of the rolling behavior is that there is an extensive variability of ways to achieve it. Infants have multiple rolling strategies, ranging from two stationary limbs to none [4]. We compare our model's support and movement patters with the values reported in [4], [8], [11]. In those studies, infants are classified into two age groups: "younger" (5 and 7 months old) and "older" (8 to 10 months old). To match these age groups, we compare with the MIMo models trained with 6 and 9-month-old embodiments.

The results of this comparison are shown in Fig. 3. We observe that MIMo, like the real infants, employs a variety of different rolling strategies. While the actual distributions of coordination patterns do no match exactly, some qualitative similarities can be identified. In particular, both the real and simulated infants expand their repertoire of support patterns between the younger and older groups.

C. Adaptive behavior within a single run

Every single model has an absolute laterality index of 1 (see [4]), meaning every trained MIMo turns either to the left or to the right, but never in both directions. This raises the question of whether the variability of coordination patterns described before is only a result of random model initializations rather than due to MIMo actually being able to learn adaptive behaviors. To address this, we arbitrarily select an individual training instance from a 9-month-old embodiment and inspect its support and movement patterns. The resulting distributions are shown in Fig. 4.a. We find that, while this model has a marked preference for rolling with two stationary limbs according to pattern A, it also shows considerable amounts of other patterns, including over 5% of rolls with no stationary limbs. A repeated analysis across multiple individual training instances shows that 85% of models roll with at least 2 patterns, 41% with at least 3 patterns, and 20% with at least 20 patterns.

To get a better understanding of this observation, we visualize the actuation commands used by a single model for each of the patterns, as shown in Fig. 4.b, where we compare the control of the torso and ipsilateral leg. A comparison of the activations, e.g., from pattern movements B and D, shows that MIMo can adapt the relative control of his leg and torso to different scenarios. Interestingly, despite there being no *a priori* incentive to learn different movements, this model executes a wide variety of coordinated actuation control sequences that explain the variability in movement patterns. Recall that the only difference from one test episode to the other is the initial condition, which results from a mild jittering of MIMo's limbs prior to the start of the episode. Nonetheless, MIMo's policy allows him to adapt to these changing scenarios.

IV. DISCUSSION

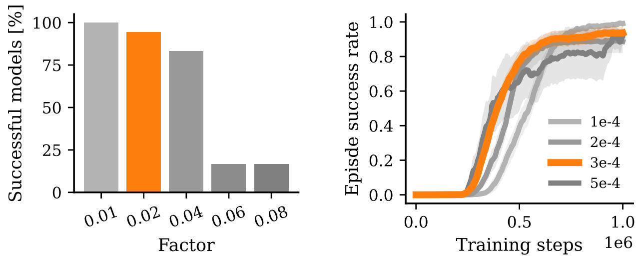
We have shown that MIMo, a virtual multimodal infant embodiment, can be trained to roll via reinforcement learning. The variability of coordination patterns exhibited matches qualitatively behavioral results [4], [8], [11], and is explained by different combinations of actuation commands. While MIMo's learning mechanism is only a crude approximation of that of a real infant, the distribution of strategies he finds to roll offers an encouraging outlook on the interpretation of the results obtained from these simulations.

A closer match between our results and the reports from real infants could be achieved with increased input about resting poses. For example, it has been reported that sleep preferences can affect the latency and distributions of rolling strategies [21]. In the absence of stronger priors, the variability displayed by our model is nonetheless quite informative. MIMo is solely attempting to maximize rewards by rotating his torso. One could expect that, in spite of the random initial conditions, different training iterations would discover and converge to a same policy, yielding a single coordination pattern. That is not the case. Thus, we speculate that the different movement sequences discovered by MIMo and by the real infants provide near-optimal solutions to rolling under variable conditions.

Our main contribution has been the identification of the role of the body in the emergence of the rolling behavior. We trained the same algorithms for the same amount of time but on different embodiments, corresponding to 1, 3, 6, and 9 months of age. The results showed a clear developmental progression with higher success rates and faster movements. Crucially, by standardizing the learning conditions, we were able to identify that physical strength is a critical component of this behavior. In line with this finding, behavioral experiments have shown that head control is a precursor to rolling [3].

There are a number of extensions that can follow this work, in particular aiming to increase the proximity of MIMo's behavior to real infants. Firstly, we will extend our experiments to more conditions, including inclined environments [10] as well as prone-to-supine rolls. Preliminary results (not shown) anticipate that this behavior may be more difficult to learn than the supine-to-prone rolls. To encourage exploration during training, we plan to incorporate curiosity-driven intrinsic motivations [22] as well as temporally auto-correlated action noise. Future experiments must also consider the contributions of vision and touch. We anticipate that these improvements can yield even more realistic rolling movements. A possible explanation for the difference in rolling times between real infants and MIMo might be that real infants have to deal with substantial conduction delays of sensory and motor signals from and to the periphery. Such delays are known to cause instabilities and oscillations in control systems. We speculate that young infants' relatively slow movements may help to avoid such problems. Modeling sensorimotor delays is possible within the MIMo ecosystem [16].

Rolling is one of the first motor milestones of full-body coordinated control and its study can provide unique insights about early development. However, acquiring data from rolling infants is a major challenge that slows down progress. Alternative methods, including video recordings [23] and subsequent motion retargeting analysis, can facilitate this process. As shown in our work, embodied computational simulations can provide a complementary approach without



(a) Penalization factor.

(b) Learning rate.

Fig. 5: Training performance for different hyperparameters. Highlighted in orange are the values chosen for the main experiments.

many of the limitations of behavioral research. The use of artificial agents opens the door for creative experimentation *in silico* and allows for large scale studies of a wide variety of conditions, including atypical physical and cognitive development.

APPENDIX

COORDINATION PATTERNS

The categorization of support patterns used by [4], [8] is based on the stationary limbs. The accronyms indicate:

TS Two stationary limbs.
IA One stationary limb (IA).
IAb One stationary limb (IA) with IL moving backwards.
IL One stationary limb (IL).
ILb One stationary limb (IL) with IA moving backwards.
NS No stationary limbs.

The refined categorization of movement patterns proposed by Kobayashi et al. [4] is as follows:

A Two stationary limbs (IA and IL) with the other two limbs (CA and CL) moving synchronously.
B Two stationary limbs (IA and IL) with the CA moving synchronously and the CL following.
C One stationary limb (IA) with the other three limbs moving synchronously.
D One stationary limb (IA) with the IL and CA moving synchronously and the CL following.
E One stationary limb (IL) with the IA and CA moving synchronously and the CL following.
F No stationary limbs with four limbs moving synchronously.

Training hyperparameters

The choice of training hyperparameters used in this study was informed by an initial sweep search. We compared different penalization factors α (see Section II-D) and learning rates using a 9-month-old MIMo trained with supine-to-prone roll overs. The results are shown in Fig. 5. We selected α as the highest value yielding more than 90% successful models ($\alpha = 0.02$ resulted in 94% successful models). We also selected a learning rate of $3 \cdot 10^{-4}$ since it provided the best balance between training speed and end performance. For all other model hyperparameters we used the defaults from the Stable-Baselines3 implementation of PPO.

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